

TIC 70762350

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.791085 d
Epoch (t0) 1355.3548 BTJD
Duration 0.1200 d
Depth 7.147 % (71,472 ppm)
SNR 66.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 7 (gate >= 3)
Robust depth ratio 1.04 (gate >= 0.50)
Robust depth sigma 51.1 (gate >= 7.0)
Odd/even z-score 0.70 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 2 light-curve dip in this sector: 43.1 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 15.20 (gate >= 5.0) Compactness 0.93 (gate >= 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 70762352
2.01 px away, dTmag +2.45 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

